



Excel From Zero

Learn.
Practice.
Apply.
Excel!

EXCEL

HANDWRITTEN NOTES

15. Excel Interview Questions
+ Practical Cheat Sheet

Beginner Friendly • Interview Focused • Job Ready

- ✓ Formulas
- ✓ Functions
- ✓ Data Analysis
- ✓ Charts
- ✓ Pivot Tables
- ✓ And More!



Step by Step
Concepts

Practical
Examples

Interview
Ready



From Basics to Brilliance,
One Formula at a Time.



Understand
Concepts



Practice
Daily



Analyze
Data



Crack
Interviews

1. Excel Introduction + Interface

★ What is Microsoft Excel?

Microsoft Excel is a powerful spreadsheet program developed by Microsoft. It allows us to organise, calculate, analyse and visualise data in a tabular format using rows and columns.



Think of Excel as a digital notebook where you can store numbers, text and formulas and perform powerful calculations within seconds.

★ Features of Excel

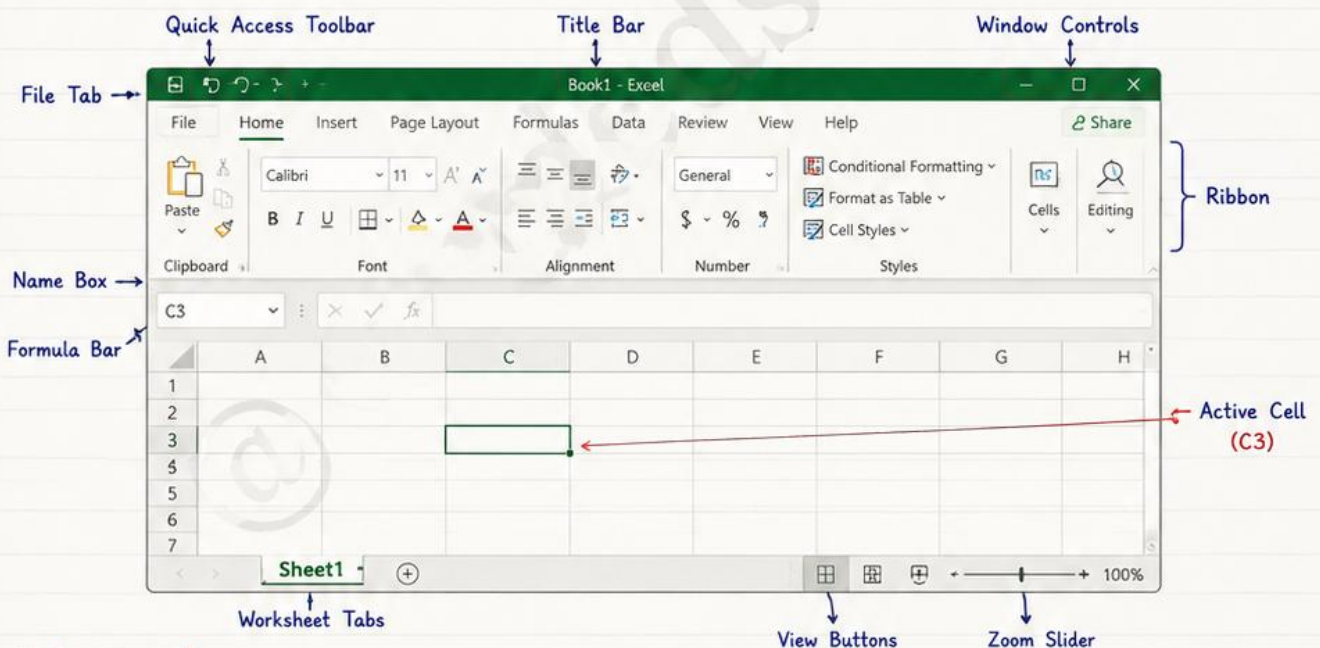
- ✓ Easy data entry and organization
- ✓ Powerful calculations with formulas
- ✓ Data analysis and summarization
- ✓ Charts and reports
- ✓ Automation with functions
- ✓ Saves time and increases accuracy

★ Common Uses

- Student marksheets
- Attendance sheets
- Monthly budgets
- Salary calculations
- Bills and invoices
- Charts and reports



★ Excel Interface (Overview)



★ Important Terms

- ① **Workbook** : A file created in Excel (e.g., Book1.xlsx).
- ② **Worksheet** : A single sheet inside a workbook (e.g., Sheet1).
- ③ **Cell** : The basic unit where data is entered. Example: C3.
- ④ **Cell Address** : Combination of column letter and row number. Example: C3.
- ⑤ **Range** : A group of cells. Example: A1:D5.

Example

Cell Address: C3 3
 ↑ ↑
 Column Row

Shortcut Keys (Remember!)

Ctrl + N : New Workbook



Ctrl + O : Open Workbook



Ctrl + S : Save Workbook



Ctrl + P : Print Workbook



Key Takeaway

Excel helps you turn raw data into useful information.



2. Worksheets + Cells + Data Entry

★ What is a Worksheet?

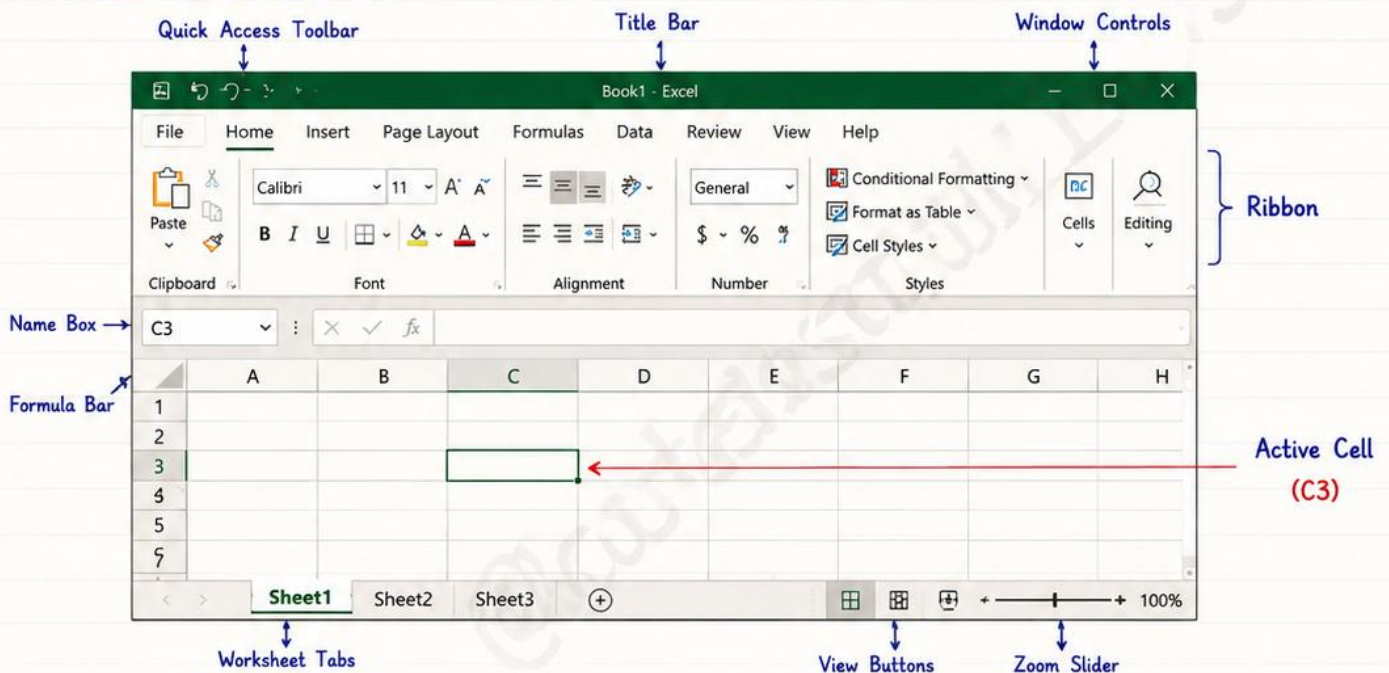
A worksheet is a single page (sheet) inside a workbook. We can store and organise data in rows and columns.



Think

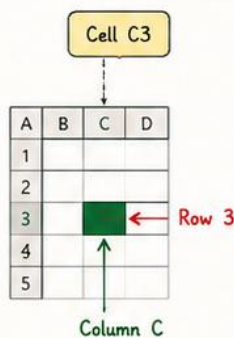
A worksheet is like a digital grid made of rows and columns where data lives.

★ Excel Window Overview



★ Rows, Columns & Cells

- Columns are vertical and identified by letters (A, B, C, ...)
- Rows are horizontal and identified by numbers (1, 2, 3, ...)
- A Cell is the intersection of a row and a column. Example: C3



★ Data Entry in Excel

- Select a cell using mouse or keyboard.
- Type the data directly in the cell.
- Press Enter to move to the cell below. Press Tab to move to the cell on the right.
- Data appears in both the cell and in the Formula Bar.
- We can edit data by selecting the cell and typing again.

Data Types

- Text → "Hello"
- Number → 1234
- Date → 25-05-2025
- Time → 10:30 AM
- Formula → =SUM(A1:A5)
- Logical → TRUE / FALSE



Tips

Active cell has a bold border.

- ✓ Use arrow keys to move quickly.
- ✓ Double click the cell border to AutoFit column width.
- ✓ Always check the data in the Formula Bar.

Shortcut Keys (Remember!)

- | | | | |
|-------------|----------------------|-------------|------------------------|
| Enter | : Move to cell below | Ctrl + Home | : Go to cell A1 |
| Tab | : Move to cell right | Ctrl + End | : Go to last used cell |
| Shift + Tab | : Move to cell left | F2 | : Edit active cell |

3. Formatting + Number Formats

★ What is Formatting?

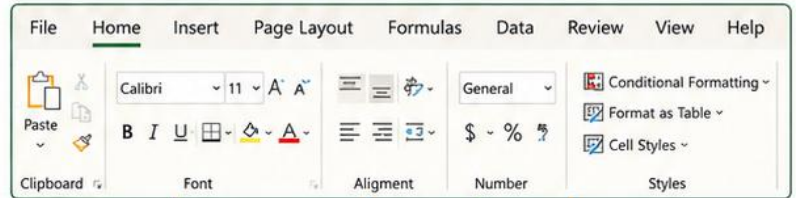
Formatting means changing the appearance of data to make it easy to read, understand and look professional.



Think

Good formatting makes your data clear and attractive!

★ Formatting Tools in Excel (Home Tab)



Font

Change font name, size, style, color.

Alignment

Align text left, center, right or justify.

Number

Apply number formats like Currency, %, Date etc.

Styles

Use cell styles, conditional formatting & format as table.

★ Font Formatting

- Change font type and size
- Make text Bold (B), Italic (I), Underline (U)
- Change font color
- Fill cell color (shading)
- Add borders to cells

★ Alignment Options

- Align Left
- Center
- Align Right
- Top Align
- Middle Align
- Bottom Align
- Wrap Text - Show all content in a cell
- Merge & Center - Combine cells

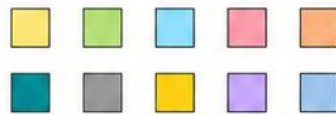
★ Borders

Apply borders to cells or ranges.

- All Borders
- Left Border
- Outside Borders
- Right Border
- Top Border
- Bottom Border
- Diagonal Border

★ Fill / Cell Shading

Fill color helps highlight important data.



More Colors...

★ Number Formats

Number format changes the way numbers or data are displayed without changing the actual value.

Format Type	Example	What it shows
General	1234	Default format
Number	1234.00	Number with decimal places
Currency	₹ 1,234.00	Number with currency symbol
Accounting	₹ 1,234.00	Currency aligned with symbols
Percentage	85%	Number in percentage
Date	25-05-2025	Date in selected style
Time	10:30 AM	Time in selected style
Text	00123	Number stored as text
Custom	#,##0.00	Create your own format

★ Example

Before Formatting

Product	Sales	Date
laptop	25000	25-05-2025
mobile	15000	26-05-2025
tablet	8000	27-05-2025

After Formatting

Product	Sales (₹)	Date
Laptop	₹ 25,000.00	25-May-2025
Mobile	₹ 15,000.00	26-May-2025
Tablet	₹ 8,000.00	27-May-2025



Tips

- ✓ Use bold headings for better readability.
- ✓ Use cell colors to highlight important information.
- ✓ Use number formats instead of adding symbols manually.
- ✓ Use borders to separate and organize data clearly.

4. Basic Formulas + Operators

★ What is a Formula?

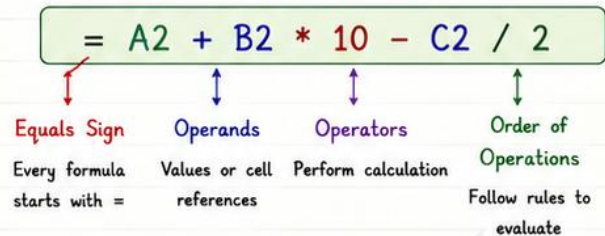
A formula is an equation that performs calculations using values in your worksheet. In Excel, every formula starts with an equals sign (=).



Think

Formulas help Excel do the hard work for you automatically!

★ Anatomy of a Formula



★ Basic Operators in Excel

Operator	Name	Example	Meaning
+	Addition	=A1+B1	Adds values
-	Subtraction	=A1-B1	Subtracts values
*	Multiplication	=A1*B1	Multiplies values
/	Division	=A1/B1	Divides values
^	Exponent	=A1^2	Raises to a power
%	Percent	=A1*10%	Percent of a value

★ Examples

Formula	Description	Example Result
=A1+B1	Add values in A1 and B1	(25 + 10 = 35)
=A1-B1	Subtract B1 from A1	(25 - 10 = 15)
=A1*B1	Multiply A1 by B1	(25 * 10 = 250)
=A1/B1	Divide A1 by B1	(25 / 10 = 2.5)
=(A1+B1)*C1	Add A1 and B1, then multiply by C1	((25+10)*2=70)
=A1+B1*C1	Multiply B1 and C1, then add A1	(25+(10*2)=45)

★ Common Formula Tips

- ✓ Always start a formula with =
- ✓ Use parentheses to control calculation order
- ✓ Double-click the fill handle to copy formulas
- ✓ Check for errors like #DIV/0! or #VALUE!
- ✓ Use meaningful cell references for better understanding

★ Order of Operations (PEMDAS)

Excel follows the PEMDAS rule to evaluate formulas.

P	Parentheses ()	Evaluate expressions inside parentheses first.
E	Exponents ^	Evaluate powers and roots.
M	Multiplication *	Do multiplication from left to right.
D	Division /	Do division from left to right.
A	Addition +	Do addition from left to right.
S	Subtraction -	Do subtraction from left to right.

★ Using Cell References

You can use cell references in formulas to make them dynamic.

	A	B	B	C
1	Item	Price	Quantity	Total
2	Pen	10	5	=B2*C2
3	Notebook	50	3	=B3*C3
4	Book	120	2	=B4*C4
5	Total			=SUM(C2:C4)



Tip

Use cell references instead of fixed values to make formulas flexible!

★ Errors You Might See

Error	Cause	Example	Solution
#DIV/0!	Division by zero	=A1/0	Check divisor value
#VALUE!	Wrong type of argument	=A1+"Text"	Use correct data type
#NAME?	Excel doesn't recognize text	=SUMM(A1:A5)	Check function spelling
#REF!	Invalid cell reference	=A1+A10 (deleted)	Fix or update reference



Key Takeaway: Formulas are the heart of Excel. Understand operators, follow PEMDAS rule, and use cell references to perform powerful calculations!



5. Essential Functions: SUM, AVERAGE, MIN, MAX, COUNT

★ Why These Functions?

These are the most commonly used Excel functions. They help us quickly calculate totals, find averages, smallest, largest and count values in a dataset.



Think

Master these five functions and 80% of your Excel work becomes easy!

★ Dataset Used in Examples

	A	B	C	D
1	Product	Jan Sales	Feb Sales	Mar Sales
2	Laptop	25000	28000	30000
3	Mobile	15000	18000	20000
4	Tablet	8000	9000	11000
5	Headphone	5000	6000	7000
6	Total/Average			

1 SUM

Adds all the numbers in a range.

Syntax

=SUM(range)

Example

=SUM(B2:B5)

Adds Jan Sales of all products
(25000 + 15000 + 8000 + 5000)

Result

53000



Tip

You can also select the cells directly and press Enter!

2 AVERAGE

Returns the average (Arithmetic mean) of numbers in a range.

Syntax

=AVERAGE(range)

Example

=AVERAGE(B2:B5)

Average of Jan Sales
(53000 / 4)

Result

13250



Tip

Average ignores blank cells but includes zeros.

3 MIN

Returns the smallest number in a range.

Syntax

=MIN(range)

Example

=MIN(B2:B5)

Smallest value in Jan Sales
(25000, 15000, 8000, 5000)

Result

5000



Tip

Useful when you want to find the minimum value.

4 MAX

Returns the largest number in a range.

Syntax

=MAX(range)

Example

=MAX(B2:B5)

Largest value in Jan Sales
(25000, 15000, 8000, 5000)

Result

25000



Tip

Useful for finding the highest value quickly.

5 COUNT

Counts the number of cells that contain numbers.

Syntax

=COUNT(value1, [value2], ...)

Example

=COUNT(B2:B5)

Counts how many numeric values are there in Jan Sales.

Result

4



Tip

COUNT ignores text and blank cells.

All Functions at a Glance

Function	What it Does	Syntax
SUM	Adds numbers	=SUM(range)
AVERAGE	Calculates average	=AVERAGE(range)
MIN	Finds smallest value	=MIN(range)
MAX	Finds largest value	=MAX(range)
COUNT	Counts numeric cells	=COUNT(range)



Key Takeaway

- ✓ Use SUM to add.
- ✓ Use AVERAGE to find the mean.
- ✓ Use MIN to find the smallest.
- ✓ Use MAX to find the largest.
- ✓ Use COUNT to count numbers.



Practice Time!

Try these:

1. =SUM(C2:C5)
2. =AVERAGE(D2:D5)
3. =MIN(D2:D5)
4. =MAX(C2:C5)
5. =COUNT(A2:A5)

Scan to Practice More



6. IF + AND + OR + Nested IF



★ Why Logical Functions?

Logical functions help us make decisions in Excel. They test conditions and return results TRUE/FALSE or specific values.



Think

Logical functions make your spreadsheets smart!
You can apply rules and get results automatically.

1 IF Function

Performs a logical test and returns one value if TRUE, another value if FALSE.

Syntax

=IF(logical_test, value_if_true, value_if_false)

Example

	A	B	C
	Marks	Result	Formula in B2
2	85	Pass	=IF(A2>=40,"Pass","Fail")
3	35	Fail	=IF(A3>=40,"Pass","Fail")
4	70	Pass	=IF(A4>=40,"Pass","Fail")



Tip

IF returns one result when the condition is TRUE and another when it is FALSE.

2 AND Function

Returns TRUE if all conditions are TRUE.

Syntax

=AND(condition1, condition2, ...)

Example

	A	B	C	D
	Name	Marks	Attendance %	Result
2	Ravi	78	85%	Pass
3	Neha	45	90%	Fail
4	Arjun	80	65%	Fail
5	Pooja	60	80%	Pass

Formula in D2 (copy down)

=IF(AND(B2>=40, C2>=75),"Pass","Fail")



Tip

All conditions must be TRUE for AND to return TRUE.

3 OR Function

Returns TRUE if at least one condition is TRUE.

Syntax

=OR(condition1, condition2, ...)

Example

	A	B	C	D
	Name	Marks	Extra Credit	Result
2	Ravi	35	Yes	Pass
3	Neha	45	No	Pass
4	Arjun	25	Yes	Pass
5	Pooja	30	No	Fail

Formula in D2 (copy down)

=IF(OR(B2>=40, C2="Yes"), "Pass", "Fail")



Tip

If any one condition is TRUE, OR returns TRUE.

4 Nested IF

Using IF inside another IF to test multiple conditions.

Syntax

=IF(condition1, value1, IF(condition2, value2, value3))

Example: Grade Calculator

	A	B	C
	Marks	Grade	Formula in B2 (copy down)
2	92	A	=IF(A2>=90,"A",IF(A2>=75,"B",IF(A2>=60,"C","D")))
3	82	B	
4	65	C	
5	45	D	



Tip

Start from the highest condition and go down. Always close brackets properly.

Logical Results in Excel

Result	Meaning
TRUE	Condition is satisfied.
FALSE	Condition is not satisfied.

Common Use Cases

- ✓ Determine Pass/Fail
- ✓ Calculate Bonus Eligibility
- ✓ Grade students
- ✓ Check multiple conditions



Scan to Practice

Quick Reference

Function	Use	Returns
IF	Single condition test	A value
AND	All conditions must be TRUE	TRUE or FALSE
OR	Any condition must be TRUE	TRUE or FALSE
Nested IF	Multiple conditions	A value





7. Text Functions

Text functions help us manipulate text data in Excel. They are very useful for cleaning, combining and extracting text.

★ Key Points

- ✓ Excel treats text inside double quotes ("").
- ✓ Text is case insensitive.
- ✓ Extra spaces can be removed using TRIM().
- ✓ Text functions do not change the original data.

Example Dataset

	A	B
	Name	Text / Value
2	" raj kumar "	
3	EXCEL is FUN	
4	excel learning	
5	India-2025	

← We will use column B for examples.

★ Most Important Text Functions

No.	Function	What it Does	Syntax	Example	Result
1	LEFT	Returns the first character(s) from a text.	=LEFT(text, num_chars)	=LEFT(B3, 5)	EXCEL
2	RIGHT	Returns the last character(s) from a text.	=RIGHT(text, num_chars)	=RIGHT(B3, 3)	FUN
3	MID	Returns a specific number of characters from the middle.	=MID(text, start_num, num_chars)	=MID(B3, 7, 2)	is
4	LEN	Returns the total number of characters (including spaces).	=LEN(text)	=LEN(B2)	13
5	TRIM	Removes extra spaces from text (keeps single space only).	=TRIM(text)	=TRIM(B2)	raj kumar
6	UPPER	Converts text to UPPERCASE.	=UPPER(text)	=UPPER(B4)	EXCEL LEARNING
7	LOWER	Converts text to lowercase.	=LOWER(text)	=LOWER(B3)	excel is fun
8	PROPER	Converts first letter of each word to Uppercase.	=PROPER(text)	=PROPER(B4)	Excel Learning
9	CONCATENATE (or &)	Joins two or more texts.	=CONCATENATE(text1, text2, ...) or text1 & text2 & ...	=B2 & "(" & B5 & ")"	raj kumar (India-2025)
10	FIND (case-sensitive)	Finds the position of text within another text.	=FIND(find_text, within_text, start_num)	=FIND("is", B3)	7

★ Other Useful Text Functions

Function	What it Does	Syntax
REPLACE	Replaces part of a text.	=REPLACE(old_text, start_num, num_chars, new_text)
SUBSTITUTE	Replaces specific text with new text.	=SUBSTITUTE(text, old_text, new_text, [instance_num])
TEXT	Converts value to text in a specific format.	=TEXT(value, format_text)

★ Quick Examples

- 1 =LEN("Excel") → 5
- 2 =UPPER("hello") → HELLO
- 3 =LOWER("WORLD") → world
- 4 =TRIM(" hi there ") → hi there
- 5 =CONCATENATE("Good", " ", "Day") → Good Day

💡 Tips

- ✓ Use TRIM() to remove extra spaces.
- ✓ Use UPPER/LOWER/PROPER for consistent text.
- ✓ Use CONCATENATE (&) to combine text.
- ✓ Use FIND instead of SEARCH when text is case-sensitive.



Key Takeaway

Text functions help you clean, extract, transform and combine text to make your data accurate and meaningful.

Practice makes perfect! 😊



8. Date + Time Functions

Date and Time functions help us work with dates and times in Excel. They are useful for calculations, reports and analysis.



Think

Excel stores dates as serial numbers. Understanding date functions can save you tons of time!

★ Key Points

- ✓ Dates in Excel are recognized as date values.
- ✓ Use TODAY() and NOW() to get current date/time.
- ✓ You can add or subtract days, months & years.
- ✓ Excel has dedicated functions to extract parts of a date.

★ Example Dataset

	A	B	C
	Event	Date	Time
2	Project Start	10-05-2024	09:30 AM
3	Review Meeting	25-05-2024	02:15 PM
4	Deadline	15-06-2024	05:00 PM
5	Launch	01-07-2024	11:00 AM

← We will use column B (Date) and C (Time) in examples.

★ Important Date + Time Functions

No.	Function	What it Does	Syntax	Example	Result	Notes
1	TODAY()	Returns current date.	=TODAY()	=TODAY()	17-05-2024	Updates daily.
2	NOW()	Returns current date & time.	=NOW()	=NOW()	17-05-2024 10:45 AM	Updates continuously.
3	DATE(year, month, day)	Creates a date.	=DATE(2024,5,17)	=DATE(2024,5,17)	17-05-2024	Useful for building dates.
4	TIME(hour, minute, second)	Creates a time.	=TIME(9,30,0)	=TIME(9,30,0)	09:30 AM	Use 24-hour format if needed.
5	DAY(date)	Returns day of month (1-31).	=DAY(B2)	=DAY(B2)	10	From date in B2.
6	MONTH(date)	Returns month number (1-12).	=MONTH(B2)	=MONTH(B2)	5	May = 5
7	YEAR(date)	Returns year.	=YEAR(B2)	=YEAR(B2)	2024	From date in B2.
8	HOUR(time)	Returns hour (0-23).	=HOUR(C2)	=HOUR(C2)	9	From time in C2.
9	MINUTE(time)	Returns minute (0-59).	=MINUTE(C2)	=MINUTE(C2)	30	From time in C2.
10	SECOND(time)	Returns second (0-59).	=SECOND(C2)	=SECOND(C2)	0	From time in C2.
11	DATEDIF(start, end, unit)	Calculates difference between two dates.	=DATEDIF(B2,TODAY(),"d")	=DATEDIF(B2,TODAY(),"d")	7	Units: "d" days, "m" months, "y" years
12	EDATE(start_date, months)	Adds months to a date.	=EDATE(B2,1)	=EDATE(B2,1)	10-06-2024	Add 1 month.

★ Add / Subtract Dates

What it Does	Example	Result
Add 7 days	=B2+7	17-05-2024
Subtract 10 days	=B2-10	30-04-2024
Add 1 month	=EDATE(B2,1)	10-06-2024
Add 3 months	=EDATE(B2,3)	10-08-2024
Add 1 year	=EDATE(B2,12)	10-05-2025

★ Common DATEDIF Units

Unit	Meaning
"d"	Difference in days
"m"	Difference in months
"y"	Difference in years
"ym"	Months excluding years
"md"	Days excluding months and years
"yd"	Days excluding years

★ Quick Examples

- ① Days passed since start date:
=DATEDIF(B2, TODAY(), "d")
- ② Months left till deadline:
=DATEDIF(TODAY(), B4, "m")
- ③ Extract year from date in B3:
=YEAR(B3)
- ④ Extract time (hour) from C4:
=HOUR(C4)



Tips

- ✓ Use today() for current date (no manual entry).
- ✓ Use eomonth(start_date, months) to get last day of month.
Example: =EOMONTH(B2,0) → Last day of month of B2.



Key Takeaway

Date & Time functions help you analyze, calculate and present data based on time intelligently. Master them to save time and avoid errors!

Practice Time!

Try calculating the number of days between Project Start (B2) and Launch (B5) using DATEDIF.

Formula: _____

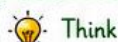


9. Lookup Functions: VLOOKUP, HLOOKUP, XLOOKUP

★ Why Lookup Functions?

Lookup functions help us search for a value in a table/range and return a related value.

They save time and reduce manual work.



Think

Lookup = Find something and bring the correct information from somewhere else!

Example Dataset

Emp ID	Name	Department	Salary
101	Ravi	Sales	45000
102	Neha	HR	52000
103	Arjun	IT	60000
104	Pooja	Finance	55000
105	Karan	Marketing	47000

We will use this table for examples.

1 VLOOKUP (Vertical Lookup)

Looks for a value in the first column of a table and returns a value from the same row in a specified column.

Syntax

`=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`

- **lookup_value** : value to search
- **table_array** : range of cells
- **col_index_num** : column number to return value
- **range_lookup** : TRUE (approx) or FALSE (exact)

Example

Find salary of Emp ID 103

`=VLOOKUP(103, A2:D6, 4, FALSE)`

Result

60000



Tip

- ✓ Must lookup in the first column.
- ✓ For exact match, use FALSE.
- ✓ For approximate match (sorted data), use TRUE.

2 HLOOKUP (Horizontal Lookup)

Looks for a value in the first row of a table and returns a value from the same column in a specified row.

Syntax

`=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])`

- **lookup_value** : value to search in first row
- **table_array** : range of cells
- **row_index_num** : row number to return value
- **range_lookup** : TRUE (approx) or FALSE (exact)

Example

Find Department of Emp ID 104

`=HLOOKUP(104, A1:D2, 2, FALSE)`

Result

Finance



Tip

- ✓ Must lookup in the first row.
- ✓ For exact match, use FALSE.
- ✓ For approximate match (sorted data), use TRUE.

Example Table for HLOOKUP

Emp ID	101	102	103	104	105
Department	Sales	HR	IT	Finance	Marketing

3 XLOOKUP (Modern Lookup)

Looks for a value in a range and returns a value from another range. More powerful than VLOOKUP/HLOOKUP.

Syntax

`=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`

- **lookup_value** : value to search
- **lookup_array** : range to search in
- **return_array** : range to return from
- **if_not_found** : value if not found
- **match_mode** : 0=exact, -1=next smaller, 1=next larger
- **search_mode** : 1=first to last (default), -1=last to first, 2=binary search (asc), -2=binary search (desc)

Example

Find Name of Emp ID 105

`=XLOOKUP(105, A2:A6, B2:B6, "Not Found")`

Result

Karan



Tip

- ✓ Works in any direction (left to right, right to left, top to bottom).
- ✓ No need to worry about column position.
- ✓ Returns "Not Found" if value is missing.

★ VLOOKUP vs HLOOKUP vs XLOOKUP

Feature	VLOOKUP	HLOOKUP	XLOOKUP
Search Direction	Vertical (Top to Bottom)	Horizontal (Left to Right)	Any Direction
Lookup Column/Row	First Column	First Row	Any Column/Row
Return Flexibility	Right side only	Below only	Any column/row
Exact/Approx Match	Yes	Yes	Yes + More options
Ease of Use	Moderate	Moderate	Easy & Powerful
Available In	Old & New Excel	Old & New Excel	Excel 365 / 2021+

Quick Examples

- 1 Salary of 102 (VLOOKUP)
`=VLOOKUP(102, A2:D6, 4, FALSE)` → 52000
- 2 Department of 101 (HLOOKUP)
`=HLOOKUP(101, A1:D2, 2, FALSE)` → Sales
- 3 Name of 103 (XLOOKUP)
`=XLOOKUP(103, A2:A6, B2:B6, "Not Found")` → Arjun
- 4 Salary of 999 (XLOOKUP with if_not_found)
`=XLOOKUP(999, A2:A6, D2:D6, "Not Found")` → Not Found



Key Takeaway

Use VLOOKUP when working with older Excel and data is vertical.
Use HLOOKUP for horizontal tables.
Use XLOOKUP for maximum flexibility and better results!

Practice Time!

Try these on the example dataset:

1. Find Name of Emp ID 104 using XLOOKUP
2. Find Department of Emp ID 105 using VLOOKUP
3. Find Salary of Emp ID 101 using HLOOKUP

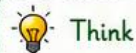




10. INDEX + MATCH

★ Why INDEX + MATCH?

INDEX and MATCH are powerful functions that work together to look up values in a table. They are more flexible than VLOOKUP and HLOOKUP.



Think

INDEX returns a value from a range based on row and column number. MATCH returns the position of a value in a row or column.

Example Dataset

	A	B	C	D
1	Emp ID	Name	Department	Salary
2	101	Ravi	Sales	45000
3	102	Neha	HR	52000
4	103	Arjun	IT	60000
5	104	Pooja	Finance	55000
6	105	Karan	Marketing	47000

We will use this table for all examples.

1 INDEX Function

Returns the value of a cell in a range based on the row number and column number you specify.

Syntax

`=INDEX(array, row_num, [column_num])`

Example

Get the Salary of Emp ID 103
(Row 4 in dataset, Column 4)

`=INDEX(A2:D6, 4, 4)`

Result

60000



Tip

- ✓ row_num = row position in the range.
- ✓ column_num = column position in the range.
- ✓ If column_num is omitted, it returns the entire row.

How it works?

- Array : A2:D6 (the whole table except header)
- Row_num : 4 → 4th row (Arjun)
- Column_num : 4 → 4th column (Salary)
- Returns the value at that position → 60000

2 MATCH Function

Returns the position of a value in a row or column range.

Syntax

`=MATCH(lookup_value, lookup_array, [match_type])`

Example

Find the position of Emp ID 103 in the Emp ID column (A2:A6)

`=MATCH(103, A2:A6, 0)`

Result

3

(103 is in 3rd position)



Tip

- ✓ match_type 0 is most commonly used.
- ✓ MATCH returns the position, not the value.

match_type:

- 0 → Exact match (default)
- 1 → Less than (sorted ascending)
- 1 → Greater than (sorted descending)

3 INDEX + MATCH Together

Use MATCH to find the position of the lookup value, and INDEX to return the value from that position.

Syntax

`=INDEX(return_range, MATCH(lookup_value, lookup_array, 0), [column_num])`

Example

Get the Salary of Emp ID 104 (Pooja)

`=INDEX(D2:D6, MATCH(104, A2:A6, 0), 1)`

Return range
(Salary column)

Find position of 104
in Emp ID column
(Result = 4)

Column number in
return_range (1st
column since it's
only one column)

Result

55000

Why use INDEX + MATCH?

- ✓ Works in any direction (left, right, up, down)
- ✓ Not affected by column/row changes
- ✓ More flexible than VLOOKUP/HLOOKUP
- ✓ Handles multiple criteria easily

INDEX + MATCH vs VLOOKUP

Feature	VLOOKUP	INDEX + MATCH
Search Direction	Left to Right only	Any direction
Lookup Column	First column only	Any column
Insert/Delete Columns	Breaks formula	Does not break
Performance	Slower on large data	Faster on large data
Flexibility	Less flexible	Highly flexible

More Examples

- 1 Get Name of Emp ID 102
`=INDEX(B2:B6, MATCH(102, A2:A6, 0), 1) → Neha`
- 2 Get Department of Emp ID 105
`=INDEX(C2:C6, MATCH(105, A2:A6, 0), 1) → Marketing`
- 3 Get Salary of Emp ID 101
`=INDEX(D2:D6, MATCH(101, A2:A6, 0), 1) → 45000`

Practice Time!

Try these on the dataset above:

1. Get Name of Emp ID 103
2. Get Department of Emp ID 102
3. Get Salary of Emp ID 105
4. Get Name of Emp ID 999
(Use IFERROR to handle not found)

Tip: Wrap INDEX+MATCH with IFERROR for better error handling.



Key Takeaway

MATCH finds the position.
INDEX returns the value.
Together, they are unbeatable!



Pro Tip

Always use match_type = 0 for exact match unless you need approximate match.



Remember

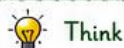
INDEX returns the value at a position.
MATCH returns the position of a value.



11. Sorting + Filtering + Conditional Formatting

★ Why are these important?

They help you organize data, focus on what matters and highlight important information visually for quick insights.

**Think**

Clean data → Better analysis
Better analysis → Better decisions!

Example Dataset

A	B	C	D	E
ID	Name	Department	Salary	Sales
101	Ravi	Sales	45000	120000
102	Neha	HR	52000	95000
103	Arjun	IT	60000	150000
104	Pooja	Finance	55000	110000
106	Karan	Marketing	47000	80000

1 SORTING

Sorting arranges your data in a specific order (A to Z, Z to A, Smallest to Largest, etc.).

Steps:

1. Select the data.
2. Go to Data tab → Sort.
3. Choose the column to sort by.
4. Choose the order (A to Z / Z to A / Smallest to Largest / Largest to Smallest).

Example: Sort by Salary (Largest to Smallest)

ID	Name	Department	Salary	Sales
103	Arjun	IT	60000	150000
104	Pooja	Finance	55000	110000
102	Neha	HR	52000	95000
106	Sneha	Sales	49000	135000
105	Karan	Marketing	47000	80000
101	Ravi	Sales	45000	120000

**Tips**

- ✓ You can sort by multiple columns (custom sort).
- ✓ Sorting changes the order of rows.

2 FILTERING

Filtering hides rows that don't meet the criteria and shows only the data you want.

Steps:

1. Select the data.
2. Go to Data tab → Filter (funnel icon).
3. Click the drop-down arrow in the column header.
4. Choose the values or set conditions.

Example 1: Show only Department = Sales

ID	Name	Department	Salary	Sales
101	Ravi	Sales	45000	120000
106	Sneha	Sales	49000	135000

Example 2: Show Employees with Salary > 50000

ID	Name	Department	Salary	Sales
102	Neha	HR	52000	95000
103	Arjun	IT	60000	150000
104	Pooja	Finance	55000	110000

**Tips**

- ✓ You can filter text, numbers, dates and colors.
- ✓ Use Text Filters, Number Filters for conditions.
- ✓ Click the filter icon again to clear or use Clear Filter.

3 CONDITIONAL FORMATTING

Conditional Formatting highlights cells automatically based on rules or conditions.

Steps:

1. Select the cells.
2. Go to Home tab → Conditional Formatting
3. Choose a rule or create a new one.
4. Set the conditions and formatting.

Examples

Rule	Example	Result
Highlight Cells Rules	Greater Than 50000 (in Salary)	Salaries > 50000 highlighted in green
Top/Bottom Rules	Top 3 Sales	Top 3 sales values highlighted
Data Bars	Show Data Bars in Sales	
Color Scales	3-Color Scale in Salary	
Icon Sets	3 Symbols in Sales	

**Tips**

- ✓ Makes important data stand out.
- ✓ Color scales show high, medium, low values.
- ✓ Icon sets give a quick visual signal.
- ✓ Use Clear Rules to remove formatting.

QUICK REFERENCE

Sorting

- A to Z (Text)
- Z to A (Text)
- Smallest to Largest
- Largest to Smallest
- Custom Sort

Filtering



- Text Filters
- Number Filters
- Date Filters
- Filter by Color
- Clear Filter

Conditional Formatting



- Highlight Cells Rules
- Top/Bottom Rules
- Data Bars
- Color Scales
- Icon Sets
- New Rule / Clear Rules

Before vs After

Before (Unsorted)

ID	Name	Salary
101	Ravi	45000
102	Neha	52000
103	Arjun	60000
104	Pooja	55000
105	Karan	47000
106	Sneha	49000



After (Sorted by Salary ↓)

ID	Name	Salary
103	Arjun	60000
104	Pooja	55000
102	Neha	52000
106	Sneha	49000
105	Karan	47000
101	Ravi	45000



Key Takeaway

Sort to organize.
Filter to focus.
Conditional Formatting to highlight.
These 3 make your data powerful!



Pro Tip

Combine all three!
Sort your data, filter what you need,
and highlight what matters most.

Practice Time!

1. Sort the data by Sales (Largest to Smallest)
2. Filter only Department = IT
3. Highlight Salaries > 50000 using Conditional Formatting





12. Tables + Data Validation

★ Why are these important?

- Tables organize your data and make it easy to analyze and work with.
- Data Validation helps you control what users can enter, reducing errors and keeping your data clean.



Think

Organized data + Controlled input
= Accurate reports, better decisions
and less rework!

Example Dataset

ID	Name	Product	Category	Qty	Price	Total
1001	Aditi	Laptop	Electronics	2	45000	90000
1002	Rohan	Chair	Furniture	4	2500	10000
1003	Neha	Pen	Stationery	10	20	200
1004	Karan	Monitor	Electronics	1	12000	12000
1005	Pooja	Table	Furniture	1	8000	8000

1 TABLES

How to Create a Table?

Excel Tables convert a range of cells into a structured data set with powerful features.

1. Select your data range.
2. Press Ctrl + T
(or Go to Insert → Table)
3. Check "My table has headers".
4. Click OK.

Example Table

ID	Name	Product	Category	Qty	Price	Total
1001	Aditi	Laptop	Electronics	2	45000	90000
1002	Rohan	Chair	Furniture	4	2500	10000
1003	Neha	Pen	Stationery	10	20	200
1004	Karan	Monitor	Electronics	1	12000	12000
1005	Pooja	Table	Furniture	1	8000	8000



Benefits of Tables

- ✓ Automatic filtering
- ✓ Banding (row shading)
- ✓ Easy sorting
- ✓ Structured references (e.g., =Table1[Total])
- ✓ Auto-expanding when you add new data
- ✓ Summarized totals row

Table Tools (Design Tab)



Table Name

Change table name for easy reference (e.g., SalesTable)



Summarized with Total Row

Add totals, averages, counts, min, max, etc. to the bottom row.



Remove Duplicates

Remove duplicate rows based on one or more columns.



Convert to Range

Convert table back to normal range when needed.



Slicers

Add slicers for quick and interactive filtering.

2 DATA VALIDATION

Data Validation controls the type of data or values that users can enter in a cell.

How to Apply Data Validation?

1. Select the cell(s).
2. Go to Data tab → Data Validation.
3. Choose the criteria under "Allow".
4. Set options and error alert (optional).
5. Click OK.

Common "Allow" Options

Allow	Use For
Whole Number	Only whole numbers (e.g., Age)
Decimal	Numbers with decimals (e.g., Price)
List	Drop-down list of values
Date	Valid dates
Time	Valid times
Text Length	Limit text length
Custom	Custom formulas and rules



Benefits

- ✓ Prevents invalid entries
- ✓ Ensures consistent data
- ✓ Shows drop-down lists
- ✓ Improves data quality
- ✓ Saves time on checking errors

Examples of Data Validation

Example	Allow	Settings / Source	Result / How it works	Screenshot (Concept)
1. Product (Drop-down list)	List	Source: Laptop, Chair, Pen, Monitor, Table	User can only select a product from the drop-down list.	Product Laptop
2. Qty (Whole Number between 1 and 100)	Whole Number	Data: between Minimum: 1 Maximum: 100	Only whole numbers from 1 to 100 are allowed.	Qty 25
3. Discount % (Decimal between 0 and 50)	Decimal	Data: between Minimum: 0 Maximum: 50	Only decimal values between 0 and 50 are allowed.	Discount % 15.5
4. Order Date (Date in future not allowed)	Date	Data: less than or equal to Date: =TODAY()	User cannot enter a date later than today.	Order Date 17-05-2024
5. Category (List with input message)	List	Source: Electronics, Furniture, Stationery Input Message: Select a category from the list.	Shows drop-down list and friendly message when cell is selected.	Select Category Please choose a category from the drop-down list. Category Electronics



Error Alert (Optional):

You can show an error message when user enters invalid data.

Example Error



Invalid Entry!
Please enter a value between 1 and 100.



Key Takeaway

- Use Tables to organize and analyze data easily.
- Use Data Validation to control and clean data at the source.
- These features make your reports professional and error-free!



Pro Tip

- Give your tables a meaningful name.
- Use Slicers with Tables for dashboards.
- Always show input and error messages in Data Validation for better user experience.

Practice Time!

1. Convert the dataset into a Table.
2. Add Total Row and calculate SUM for Total.
3. Add a drop-down list for Category.
4. Restrict Qty between 1 and 100.
5. Restrict Price to be greater than 0.





13. Charts + Graphs

★ Why Charts?

- Charts turn numbers into visuals.
- Easy to understand.
- Helps in comparison, trends and patterns.
- Makes reports look professional.



Think

A picture is worth a thousand numbers. Charts help you see the story behind the data!

Example Dataset

Month	Sales (₹)	Cost (₹)	Profit (₹)
Jan	120000	70000	50000
Feb	150000	85000	65000
Mar	180000	100000	80000
Apr	160000	90000	70000
May	200000	110000	90000
Jun	170000	95000	75000

We will use this data for all examples.

1 Column Chart

Used to compare values across categories. Best for showing changes over time.



How to Create?

1. Select your data (including headers).
2. Go to Insert tab.
3. Click on Column Chart → 2-D Column.
4. Choose the style you like.
5. Chart is ready!

Use When:

You want to compare values across categories or time.

2 Line Chart

Used to show trends over time. Best for continuous data.



How to Create?

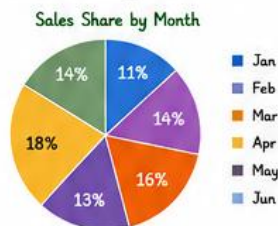
1. Select your data.
2. Go to Insert tab.
3. Click on Line Chart → 2-D Line.
4. Choose the style.
5. Chart is ready!

Use When:

You want to show trend or pattern over time.

3 Pie Chart

Used to show percentage or part-to-whole relationship. Best for few categories.



How to Create?

1. Select data (Month and Sales).
2. Go to Insert tab.
3. Click on Pie Chart → 2-D Pie.
4. Add Data Labels if needed.
5. Chart is ready!

Use When:

You want to show contribution or share in total.

4 Bar Chart

Similar to Column chart but horizontal. Good for comparing long category names.



How to Create?

1. Select your data.
2. Go to Insert tab.
3. Click on Bar Chart → 2-D Bar.
4. Choose the style.
5. Chart is ready!

Use When:

You want to compare categories with long names.

5 Doughnut Chart

Similar to Pie chart but with a hole in the center.



Use When:

You want to show part-to-whole but in a modern style.

6 Combination Chart

Used to compare two or more data series with different scales (e.g., Sales in bars and Profit in line).



How to Create?

1. Select all data.
2. Go to Insert tab.
3. Click on Combo Chart.
4. Choose Column for Sales and Line for Profit.
5. Click OK.

Popular Chart Types at a Glance

Chart Type	Best For	Example Use	Highlights
Column Chart	Comparing values	Monthly Sales comparison	Simple and easy to read
Line Chart	Showing trends	Sales trend over months	Shows ups and downs
Pie Chart	Part-to-whole	Market share	Use for few categories (max 6-7)
Bar Chart	Comparing long categories	Top products by sales	Horizontal bars, easy to read
Doughnut Chart	Part-to-whole (modern look)	Expense breakup	Similar to Pie, with center hole
Area Chart	Showing volume over time	Website traffic over months	Filled area shows volume
Scatter Chart	Show relationship	Ad spend vs Sales	Find correlation between two values
Combo Chart	Compare multiple data	Sales (bars) vs Profit (line)	Best for different scales



Tips for Better Charts

- ✓ Always add Chart Title.
- ✓ Use Axis Titles.
- ✓ Keep it simple, avoid clutter.
- ✓ Use proper colors.
- ✓ Add Data Labels when needed.
- ✓ Don't use 3D unless necessary.
- ✓ Keep font size readable.
- ✓ Choose the right chart for the data.



Key Takeaway

- Charts help in visual storytelling.
- Choose the right chart for the right data.
- Keep your charts simple, clear and meaningful.
- Good charts = Better insights!



Pro Tip

- Update your chart automatically when data changes.
- Use Table (Ctrl + T) for dynamic charts.
- Use Slicers for interactive charts.

Practice Time!

Create the following charts using the dataset:

1. Column Chart for Sales
2. Line Chart for Profit
3. Pie Chart for Sales Share
4. Bar Chart for Cost
5. Combo Chart for Sales vs Profit





14. Pivot Tables + Pivot Charts

★ Why are they important?

- Summarize large amounts of data quickly.
- Find insights without complex formulas.
- Easily change the view (drag & drop).
- Create dynamic charts that update automatically.



Think

A Pivot Table is a flexible summary of your data.

A Pivot Chart is a visual representation of the Pivot Table.

Example Dataset

Date	Region	Product	Sales	Quantity
01-Jan	North	Laptop	1200	2
03-Jan	South	Phone	800	3
05-Jan	East	Laptop	1500	1
07-Jan	West	Tablet	600	4
09-Jan	North	Phone	950	2
11-Jan	South	Laptop	1800	1
13-Jan	East	Tablet	750	3
15-Jan	West	Phone	1100	2

We will use this data for our Pivot Table and Pivot Chart examples.

1 Create a Pivot Table

Steps:

1. Select your data range.
2. Go to Insert → PivotTable.
3. Choose the table/range and location (new or existing worksheet).
4. Click OK.

PivotTable Fields

Drag fields to the areas below:

☒ Date ☒ Region ☒ Product ☒ Sales ☐ Quantity	Filters 	Columns Product
	Rows Region	Values Sum of Sales

Pivot Table Result

Sum of Sales				
Region	Product			Grand Total
	Laptop	Phone	Tablet	
East	1500	750	750	3000
North	1200	950	0	2150
South	1800	800	0	2600
West	0	1100	600	1700
Grand Total	4500	3600	1350	9450

2 Create a Pivot Chart

Steps:

1. Click inside your Pivot Table.
2. Go to Insert → PivotChart.
3. Choose the chart type.
4. Click OK.

Pivot Chart Example



Tips

- ✓ Use Pivot Charts for quick visual insights.
- ✓ You can change the chart type (column, line, pie, etc.).
- ✓ Use slicers and timelines to filter data interactively.

3 Filter, Sort and Analyze

Use the Pivot Table tools to:

- **Filter:** Use dropdowns in Row/Column labels.
- **Sort:** Right-click a value → Sort → A to Z / Z to A.
- **Drill Down:** Double-click a value to see detail.
- **Show Values As:** % of Total, Running Total, etc.

Example: Show % of Total Sales

Sum of Sales	
Region	% of Total
North	22.7%
South	27.5%
East	31.7%
West	18.0%
Grand Total	100.0%

Use Slicers

Add slicers to filter your Pivot Table and Pivot Chart.

Region	Product
North	Laptop
South	Phone
East	Tablet
West	

4 Pivot Chart with Multiple Fields

Example: Sales by Product and Region



Pivot Chart Fields

Axis (Categories) Region	Legend (Series) Product
Values Sum of Sales	Filters

Common Options

- ✓ Change chart type (e.g., stacked, line).
- ✓ Add data labels.
- ✓ Show totals and percentages.
- ✓ Use PivotChart filters (slicers/timelines).
- ✓ Update automatically when data changes.



Key Takeaway

- Pivot Tables summarize, Pivot Charts visualize.
- Use slicers and filters for interactivity.
- Great for reports, dashboards and decision-making!



Pro Tip

- Click any cell in your data range and press **Alt + N + V + T** to create a Pivot Table quickly.
- Use **Ctrl + Z** if you make a mistake.



Practice Time!

1. Create a Pivot Table for total sales by Region and Product.
2. Add a Pivot Chart (column or bar).
3. Use a slicer for Region and show % of total sales.
4. Change the chart type to a stacked chart.





15. Excel Interview Questions + Practical Cheat Sheet

**Remember:**

Practice Daily
Work Smart
Excel = Superpower!



A Excel Interview Questions

1. Basic Excel

1. What is Excel?
2. What is the difference between a Workbook and a Worksheet?
3. What are the different data types in Excel?
4. What is the use of Autofill?
5. What is the difference between relative, absolute and mixed cell references?

2. Formulas & Functions

1. What is the difference between = and +?
2. How do you handle errors in Excel?
3. What is the use of IF, AND, OR functions?
4. Explain VLOOKUP and HLOOKUP with examples.
5. What is the difference between COUNT, COUNTA, COUNTBLANK?

3. Data Management

1. What is Data Validation? Why is it used?
2. How do you remove duplicates in Excel?
3. What is the use of Text to Columns?
4. How do you sort data by multiple criteria?
5. What is the difference between Filter and Advanced Filter?

4. Charts & Visuals

1. What are the different types of charts in Excel?
2. How do you create a Combo Chart?
3. When would you use a Pivot Chart?
4. How can you make a chart dynamic?

5. Pivot Table

1. What is a Pivot Table?
2. What are the areas in a Pivot Table?
3. How do you group dates in a Pivot Table?
4. How do you refresh a Pivot Table?
5. What is the difference between Pivot Table and Normal Table?

6. Advanced Excel

1. What is Conditional Formatting?
2. How do you use INDEX + MATCH?
3. What is the difference between VLOOKUP and XLOOKUP?
4. What is Power Query?
5. How do you link multiple sheets/workbooks?

7. Scenario / Practical

1. Given a large sales dataset, how will you create a summary report?
2. How will you find top 5 customers by sales?
3. How will you highlight duplicate values?
4. How will you track monthly sales trend?
5. How will you create a dashboard in Excel?



Tip: Show your Excel file with real examples in interviews. Practical knowledge + shortcuts = Impression!

B Practical Cheat Sheet

1. Essential Shortcuts

Ctrl + N	New Workbook
Ctrl + O	Open Workbook
Ctrl + S	Save Workbook
Ctrl + P	Print
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + C	Copy
Ctrl + V	Paste
Ctrl + X	Cut
Ctrl + F	Find
Ctrl + H	Find & Replace
Ctrl + T	Create Table
Ctrl + Arrow	Go to end of data
Ctrl + Shift + L	Apply/Remove Filter
Alt + =	AutoSum
F2	Edit Active Cell

3. Data Cleaning

- ✓ Remove Duplicates (Data Tab)
- ✓ Text to Columns (Data Tab)
- ✓ Trim spaces: =TRIM(A2)
- ✓ Proper case: =PROPER(A2)
- ✓ Remove extra spaces: =TRIM(SUBSTITUTE(A2,CHAR(160),""))

6. Pivot Table

1. Select data → Insert → PivotTable
2. Drag fields to:
 - Rows (Row Labels)
 - Columns (Column Labels)
 - Values (Sum, Count, Avg...)
 - Filters (Report Filter)
3. Use Grouping (Right click → Group)
4. Refresh PivotTable (Right click → Refresh)
5. Use Slicers for interactivity
6. PivotChart: Insert → PivotChart

9. Excel Tips

- ★ Use Tables (Ctrl + T) for dynamic ranges.
- ★ Use Ctrl + T to convert data into Table.
- ★ Freeze Panes: View → Freeze Panes
- ★ Name ranges for easy reference.
- ★ Use absolute references (\$A\$1) in formulas.
- ★ Use Format Painter to copy formatting.
- ★ Use Data Validation to control input.
- ★ Always keep backups before big changes.

2. Common Functions

Function	Use
SUM(range)	Adds numbers
AVERAGE(range)	Average of numbers
MIN(range)	Smallest value
MAX(range)	Largest value
COUNT(range)	Counts numbers
COUNTA(range)	Counts non-blank cells
COUNTBLANK(range)	Counts blank cells
IF(logical, value_if_true, value_if_false)	Logical test
AND(logical1, [logical2], ...)	All conditions TRUE?
OR(logical1, [logical2], ...)	Any condition TRUE?
SUMIF(range, criteria, [sum_range])	Sum with condition
COUNTIF(range, criteria)	Count with condition
VLOOKUP(lookup, table, col, FALSE)	Vertical lookup
INDEX(range, row, col)	Returns value
MATCH(lookup, range, 0)	Returns position
XLOOKUP(lookup, lookup_array, return_array)	Modern lookup

4. Conditional Formatting

- ✓ Highlight Cells Rules
- ✓ Top/Bottom Rules
- ✓ Data Bars
- ✓ Color Scales
- ✓ Icon Sets
- ✓ New Rule → Use Formula for custom rules

5. Charts

- ✓ Select Data → Insert → Choose Chart
- ✓ Change Chart Type anytime
- ✓ Add Chart Title, Labels, Legend
- ✓ Use Slicers for interactive charts

8. Date & Time Shortcuts

=TODAY()	Today's date
=NOW()	Current date & time
=DATE(year, month, day)	Returns date
=EOMONTH(date, n)	End of month
=DATEDIF(start, end, "d")	Difference in days
=NETWORKDAYS(start, end, holidays)	Working days
=WORKDAY(start, days, holidays)	Add working days

10. Formula Cheat Sheet (Most Used)

Formula	Example	Formula	Example
Sum	=SUM(B2:B10)	Vlookup	=VLOOKUP(E2,A2:D10,2,FALSE)
Average	=AVERAGE(B2:B10)	Hlookup	=HLOOKUP(E1,A1:H10,3,FALSE)
Count	=COUNT(B2:B10)	Index + Match	=INDEX(B2:B10,MATCH(E2,A10,0))
Count Blank	=COUNTBLANK(B2:B10)	Concatenate	=CONCAT(A2," ",B2)
Max	=MAX(B2:B10)	Left	=LEFT(A2,3)
Min	=MIN(B2:B10)	Right	=RIGHT(A2,3)
If	=IF(C2>=60,"Pass","Fail")	Mid	=MID(A2,2,4)
Sum If	=SUMIF(C2:C10,">=60",B2:B10)	Len	=LEN(A2)
Count If	=COUNTIF(C2:C10,"Pass")	Trim	=TRIM(A2)

11. Dashboard Checklist

- ✓ Clean & structured data
- ✓ Key KPIs (Top numbers)
- ✓ Charts for trends & comparisons
- ✓ Slicers / Timelines (if needed)
- ✓ Conditional Formatting for highlights
- ✓ Consistent formatting & colors
- ✓ Dynamic ranges / Tables
- ✓ Clear titles & labels

12. What Interviewers Look For

- ★ Accuracy
- ★ Problem-solving approach
- ★ Excel functions knowledge
- ★ Data handling & cleaning
- ★ Dashboard / Report building



Key Takeaway

- Practice is the key.
- Learn shortcuts and save time.
- Work with real data.
- Excel skills = Career Growth!



Pro Tip

- Solve 1 practical problem daily.
- Build your own dashboards.
- Use Excel in real-life tasks.
- Keep learning new functions.



Final Note

Master Excel step by step and you will be the go-to person for any data task! 🚀

Keep Practicing!

The more you practice, the more confident you will become.

